

OFFICIAL ENTERPRISE SYSTEM DOCUMENTATION

SWED WOOD WORK MANAGEMENT SYSTEM

Comprehensive Operational & User Guide

Dual Cloud Firestore & Local Offline Storage Architecture

SWEDSWOOD ENTERPRISE / SWEDSFREE ENTERPRISE

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Operating Currency Standard: Sierra Leone Leone (Le / SLL)

Application Release: Version 2.5 Production Build (2026 Edition)

Audience: System Administrators, Workshop Managers, Auditors & Artisans

DOCUMENT SCOPE & PURPOSE

This manual serves as the authoritative operational guide for the Swedswood Carpentry and Timber Logistics Management System. It delivers end-to-end guidance on navigating workshop workflows, managing raw timber inventory and consumable hardware, registering institutional, company, and private clients, tracking bespoke furniture commissions, generating legally compliant branded invoices and payment receipts, enforcing installment payment settlement rules, conducting financial audits, managing employee payroll in Sierra Leone Leones (Le), and executing offline backups with Zero Data Loss protection.

ZERO DATA LOSS GUARANTEE

The system operates a high-reliability dual-engine: all client dossiers, job records, inventory logs, and financial transactions persist immediately to local device cache and sync live with Google Cloud Firestore.

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CHAPTER 1: SYSTEM OVERVIEW & DUAL STORAGE ARCHITECTURE

1.1 Executive System Overview

The Swedswood Carpentry and Construction Management System is an enterprise-grade, web-based management suite purpose-built for commercial carpentry workshops, timber merchants, and bespoke furniture fabricators in Sierra Leone. The application integrates all critical aspects of workshop management into a cohesive, highly responsive user interface optimized for desktop, tablet, and mobile devices.

1.2 Zero Data Loss Dual Storage Architecture

A central engineering pillar of the system is the Zero Data Loss Guarantee. Operating in environments with frequent electrical or internet fluctuations requires resilient data persistence. The platform incorporates a synchronized two-tier persistence layer:

- **Tier 1: Browser Local Storage Cache:**

Every user keystroke, job creation, financial transaction, and material entry is instantaneously written to the local device storage cache. If connectivity is interrupted, no unsaved changes are lost, and full operational access continues smoothly.

- **Tier 2: Google Cloud Firestore Engine:**

Whenever an active internet link is present, changes are synced in real-time to the dedicated Google Cloud Firestore database. This provides real-time cross-device synchronization and permanent cloud redundancy.

KEY OPERATIONAL RULE: OFFLINE MODE READINESS

You do not need to wait for internet reconnection to log daily work, create customer records, or issue invoices. The system queues changes locally and commits them to Cloud Firestore immediately when a network handshake is re-established.

1.3 Hardware & Browser Requirements

The platform requires no specialized client installation and runs cleanly inside any modern web browser:

- **Recommended Browsers:** Google Chrome (v90+), Mozilla Firefox (v88+), Apple Safari (v14+), Microsoft Edge (v90+).
- **Display Resolution:** Optimized for high-density desktop displays (1920x1080) and fully responsive down to mobile viewports (375px).
- **Printers & Export:** Any standard A4 or Letter laser/inkjet printer, and native PDF virtual printers for direct digital dispatch.

CHAPTER 2: USER ROLES & ACCESS CONTROL (RBAC)

2.1 Role-Based Access Control Architecture

To preserve data integrity, protect sensitive financial records, and enforce operational accountability, the system employs a stringent Role-Based Access Control (RBAC) framework. Every staff member is assigned one of four designated security roles:

- **Administrator (Admin):**
Full supreme administrative authority. Manages system settings, data backup/restore, employee role assignments, database fresh start, and all workshop modules.
- **Operations Manager:**
Supervises workshop commissions, inventory intake, client registration, quote and invoice dispatch, installment clearances, and daily workforce assignments.
- **Financial Auditor:**
Specialized read-only oversight role. Has access to all financial ledgers, invoice archives, payment receipts, and the immutable Payment Audit Log, with destructive actions blocked.
- **Workshop Artisan / Staff:**
Operational ground role. Restricted to logging daily work photos, viewing assigned commissions, and recording timber/material usages.

2.2 RBAC Permission Matrix

Function / Module	Admin	Manager	Auditor	Artisan/Staff
Workshop Hub Dashboard	Full Access	Full Access	Read-Only	Restricted
Raw Timber & Inventory	Full Access	Full Access	Read-Only	Log Usage Only
Customer Directory	Full Access	Full Access	Read-Only	No Access
Jobs & Commissions	Full Access	Full Access	Read-Only	Assigned Only
Invoicing & Receipts	Full Access	Full Access	Read-Only	No Access
Payment Audit Trail	Full Access	Full Access	Audit View	No Access
Financial Cash Ledger	Full Access	Full Access	Audit View	No Access
Employee Roster & Payroll	Full Access	Full Access	Read-Only	No Access
Audit Reports & Trends	Full Access	Full Access	Full Access	No Access
Backup & Restore (Settings)	Full Access	Full Access	No Access	No Access
Database Fresh Start	Admin Only	No Access	No Access	No Access

2.3 User Authentication & Approval Workflow

1. Registration Request: New employees submit their full name, email, phone number, and desired role on the login screen.
2. Administrative Vetting: The submission enters the Pending Approval queue under the Employees tab.
3. Account Activation: An Administrator verifies credentials and activates the account, selecting appropriate security clearance.

CHAPTER 3: WORKSHOP HUB (EXECUTIVE DASHBOARD)

3.1 Real-Time Workshop KPI Metrics

The Workshop Hub serves as the primary cockpit for daily management. Upon authentication, users are presented with immediate high-level indicators reflecting the operational health of the enterprise:

- **Active Woodwork Jobs:** Count of currently running commissions across Pending, In Progress, and Quality Check states.
- **Raw Timber & Materials Stock:** Total quantity of stored lumber pieces, board feet, and hardware items, with low-stock warnings highlighted in red.
- **Total Payments Cleared:** Cumulative revenue received in Sierra Leone Leones (Le) across all recorded project deposits and installments.
- **Active Client Portfolio:** Total registered clients categorized into Corporate, Institutional, and Private accounts.

3.2 Rapid Action Shortcuts

Convenient quick-action buttons at the top of the Workshop Hub allow managers to execute frequent tasks in one click:

- **New Job Commission:** Opens the job creation modal to quote and commission a new project.
- **Log Material Intake:** Opens inventory entry to log arriving lumber trucks, board feet, or hardware.
- **Record Cash Voucher:** Logs immediate petty cash, generator fuel, or utility expenses.
- **Issue Client Invoice:** Redirects to the invoice workbench with client pre-selection.

3.3 System Health & Connectivity Indicator

The top-right header displays real-time connection status (Online / Cloud Synced or Offline Local Cache). This guarantees total transparency regarding whether cloud replication has completed.

CHAPTER 4: INVENTORY & TIMBER LOGISTICS MANAGEMENT

4.1 Inventory Categorization

Swedswood maintains strict accounting of workshop materials divided into distinct logistical classes:

- **Hardwood & Sawn Timber:** Locally sourced and imported timber species including Mahogany, Teak, Cedar, Iroko, and Plywood boards.
- **Joinery Hardware:** Heavy-duty drawer runners, ball-bearing hinges, mortise locks, handles, dowels, and structural fasteners.
- **Finishing & Chemicals:** Polyurethane sealers, wood stains, clear lacquers, sanding discs, glues, and thinners.
- **Consumables & PPE:** Safety goggles, respirators, router bits, and table-saw replacement blades.

4.2 Adding & Updating Inventory Items

1. Click "+ Add Inventory Item" within the Inventory tab.
2. Input the item name, category, timber dimensions (thickness x width x length in inches or meters).
3. Enter current stock quantity, unit of measurement (pieces, planks, packets, liters).
4. Set the purchase unit cost in Sierra Leone Leones (Le).
5. Define the Minimum Alert Threshold (e.g. 10 planks). When stock drops below this number, the system automatically flags the item.

4.3 Job Material Allocation & Wastage Management

When carpenters consume lumber on a project, materials are logged directly against the job commission ID. The system automatically deducts the quantity from central inventory and tallies the total material expenditure against the client quote to determine project profitability.

CHAPTER 5: CUSTOMER & CLIENT RELATIONSHIP MANAGEMENT

5.1 Three Client Classification Tiers

To streamline billing, terms, and tax handling, the system classifies all customer accounts into three official categories:

- **Company:** Commercial corporations, interior designers, property developers, and construction contractors.
- **Institution:** Government ministries, embassies, universities, educational boards, and non-profit organizations.
- **Private:** Individual home owners, residential clients, and private commissioners.

5.2 Client Profile Dossiers

Each client dossier retains comprehensive records including primary contact person, direct phone, WhatsApp number, email, site delivery address, historical commission count, cumulative contract value, total cash cleared, and live outstanding debt balance.

5.3 Direct Job Commissioning

Managers can initiate a job directly from a customer card by clicking "Create Job". This pre-fills customer credentials, eliminating repetitive data entry and preventing duplicate accounts.

CHAPTER 6: COMMISSION & JOB PROJECT MANAGEMENT

6.1 Commission Creation Workflow

- 1. Navigate to "Job lists" and click "+ New Commission Order".
- 2. Select an existing client or enter new client details on the fly.
- 3. Define the Project Title (e.g. "Solid Mahogany 10-Seater Executive Conference Table").
- 4. Itemize detailed specifications, dimensions, wood types, and finish requirements.
- 5. Set the Agreed Contract Quote in Sierra Leone Leones (Le).
- 6. Specify start date and client target delivery deadline.

6.2 Commission Lifecycle Stages

Stage	Operational Meaning	Financial Milestone Expectation
Pending	Quote submitted, awaiting client sign-off or deposit	Initial mobilization deposit (typically 50-70%)
In Progress	Wood selected, milling, joinery & assembly underway	Milestone progress installment (20-30%)
Quality Check	Finishing, sanding, lacquering, hardware fitting	Pre-delivery inspection & final payment clearance
Completed	Fabrication concluded, packaged for dispatch	100% contract amount fully cleared
Delivered	On-site installation and client sign-off achieved	Zero outstanding balance, project archived

6.3 Artisan & Staff Allocation

Commission cards permit the assignment of lead carpenters and assistants. Assigned personnel can review the commission specifications from their mobile devices, ensuring exact adherence to customer measurements.

CHAPTER 7: INVOICES, RECEIPTS & INSTALLMENT CLEARANCE

7.1 Commercial Invoice Generation

The Invoices & Receipts module provides complete billing capabilities. Quotes and Invoices feature official Swedswood Enterprise branding, registered tax information, banking coordinates, and custom multi-line itemization.

OFFICIAL INVOICE RULE: NO BALANCE DISPLAYED

In accordance with enterprise accounting standards, official Commercial Invoices show the complete contractual quote amount and scope of works. Outstanding debt balances are NOT displayed on invoices; payment balances are strictly managed on official Payment Receipts and Client Financial Ledgers.

7.2 Recording Installment Payments & Issuing Receipts

Clients rarely settle large custom woodwork commissions in a single upfront payment. The system provides comprehensive installment tracking:

- **Recording an Installment:**

Select the commission, click "+ Record Payment", enter the exact amount in Le, select payment method (Cash, Bank Transfer, Cheque, Mobile Money), and add notes.

- **Official Payment Receipt:**

Each captured payment generates a branded receipt displaying the Payment ID, date, method, notes, and the Captured Payment Records table.

- **Financial Settlement Breakdown:**

Every receipt automatically calculates the Total Contract Value, Cumulative Payments to Date, and the Remaining Balance Due (or displays "Le 0.00 - Fully Settled").

7.3 Safe Payment Editing & In-UI Confirmations

To avoid browser freezing inside iframe environments, all payment modifications and deletions utilize in-UI dialog modals with prominent validation error banners. Accidental deletions require explicit two-step confirmation.

7.4 Immutable Payment Modification Audit Log

Any modification or deletion of a payment installment is permanently recorded in the Payment Audit Log. The log records the timestamp, operator name, order title, action type (CREATED, UPDATED, DELETED), and the exact monetary difference. Auditors can inspect this log at any time without risking tampering.

CHAPTER 8: DAILY WORKSHOP & SITE ACTIVITY LOGS

8.1 Visual Progress Documentation

High-end carpentry requires visual proof of craftsmanship and structural integrity. The Daily Logs module enables carpenters and site managers to upload daily photographic entries documenting work stages.

- **Bench Work & Joinery:** Photos of mortise-and-tenon joints, carcass assemblies, carving details, and edge banding.
- **Finishing Stages:** Photos of wood grain filling, stain matching, primer coats, and high-gloss lacquering.
- **Site Deliveries & Installation:** Photos of kitchen cabinetry installations, architectural panelling, and final client sign-offs.

8.2 Log Search & Timeline Inspection

Logs are tagged by date, artisan, and job title. Clients inquiring about project status can be shown an accurate visual timeline proving steady progress.

CHAPTER 9: FINANCIAL ACCOUNTING & CASH FLOW LEDGERS

9.1 Financial Categorization Architecture

The Financial Ledger classifies all workshop monetary transactions into standardized revenue and expense categories:

Inwards (Revenue) Categories	Outwards (Expenditure) Categories
wood (Raw lumber sales & milling fees)	Tools and generator (Fuel, servicing, equipment purchase)
sofa (Custom upholstered furniture)	Utilities (Electricity, municipal rates, water supply)
Chair (Dining, office, and lounge seating)	Transportation (Timber haulage, site delivery fuel)
Bed (Bedframes, headboards, wardrobes)	Material Purchase (Hardware, fasteners, varnishes, glues)
Wood Construction (Roof trusses, decking)	Tools and Maintenance (Blade sharpening, machine repairs)
others (Bespoke commissions & restoration)	Cast (Foundry, metal framing & brackets)
	Salary (Artisan wages, casual labor, allowances)
	others (Sundry workshop consumables)

9.2 Net Cash Position & Audit Filtering

The ledger automatically calculates Total Inflows, Total Outflows, and Current Net Cash Surplus/Deficit. Users can filter records by date ranges, transaction types, or search keywords to reconcile cash books.

CHAPTER 10: HUMAN RESOURCES, ATTENDANCE & WAGE MANAGEMENT

10.1 Staff Roster & Artisan Profiles

The Employees module maintains profiles for master carpenters, apprentices, polishers, and machine operators. Profiles include contact telephone numbers, emergency contacts, hire dates, and role assignments.

10.2 Wage Calculations in Sierra Leone Leones (Le)

All wage computations are strictly formatted in Sierra Leone Leones (Le):

- **Base Compensation:** Configurable as Daily Rate, Hourly Wage, or Fixed Monthly Salary.
- **Overtime Multipliers:** Overtime hours logged during night shifts or weekend rush projects are calculated with configurable overtime multipliers.
- **Disciplinary & Warning Letters:**
Formal written warnings can be attached to an employee profile to ensure professional standards and punctuality across the workshop.

CHAPTER 11: AUDIT REPORTS & MONTHLY TRENDS ANALYTICS

11.1 Executive Audit Dossiers

The Audit Reports module synthesizes data from inventory, jobs, financial ledgers, and payments into clean, printable executive reports suitable for bank submissions, tax accounting, and partner reviews.

11.2 Monthly Trends & Variance Analytics

The integrated Monthly Trends section provides dynamic charts (powered by Recharts) comparing:

- **Job Completion Trajectories:** Monthly volume of finished and delivered commissions.
- **Material Cost Variances:** Fluctuations in timber and hardware purchasing costs across calendar months.
- **Profit Margin Trends:** Net revenue spread after deducting direct project materials and labor.

CHAPTER 12: SETTINGS, DATA BACKUPS & ZERO DATA LOSS PROTOCOL

12.1 Backup & Recovery Hub

Located exclusively within the Settings tab, the Backup & Recovery Hub gives authorized active users complete control over their business records:

- **Backup System Data (JSON Download):**

Exports a complete offline JSON archive containing all inventory, clients, jobs, payments, invoices, daily logs, and employee profiles. Keep this file on a secure USB drive or external drive.

- **Restore JSON File (Upload):** Restores the complete database state from a previously saved JSON snapshot in seconds.

- **Bring Back Cloud Data (Restore Till Today):**

Pulls all latest records from Google Cloud Firestore, synchronizing local browser storage with the cloud database.

SECURITY NOTE: SYSTEM FRESH START (ADMINS ONLY)

The "Clear System Data (Fresh Start)" action is strictly restricted to Administrators. It enables resetting the database for a new fiscal period or demonstration setup. All users are prompted for confirmation prior to execution.

CHAPTER 13: FREQUENTLY ASKED QUESTIONS & TROUBLESHOOTING

13.1 Common Questions

- **What happens if the internet goes offline during work?:**
You can continue working without disruption. All entries save to local storage immediately and sync with Cloud Firestore once reconnected.
- **Why doesn't the customer balance show on the printed invoice?:**
By accounting design, invoices show total quote contracts. Balances are tracked and displayed on Payment Receipts and the Customer Ledger.
- **How can I correct an accidental payment entry?:**
Navigate to Invoices & Receipts or the Customer tab, find the installment, and click the Edit icon. The change is verified in an in-UI modal and recorded in the audit trail.
- **Where can I download the latest User Manual PDF?:**
You can download the complete manual at any time from the Settings tab or the top bar by clicking "Download User Manual (PDF)".

13.2 Technical Support & Inquiries

For technical assistance, system customization, or training workshops:

- Swedswood Enterprise IT Support Desk
- 2 Sweds free Avenue, Sussex, Freetown, Sierra Leone
- Email: support@swedwoodwork.com | info@swedwoodwork.com
- Direct Helpline: +232 76 112 3344 / +232 88 554 9900
- Website: www.swedwoodwork.com